

CSE110 Fall 2025 Lab Exam - 02 Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A] SOLUTION

```
import java.util.Scanner;
public class Quiz2SetA {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Ancient Number: ");
        int num = sc.nextInt();
        System.out.print("Special digit: ");
        int N = sc.nextInt();
        int temp = num;
        int foundPos = -1;
        int temp_pos = 1;
        // Search the Number
        while (temp > 0) {
            int digit = temp % 10;
            if (digit == N) {
                foundPos = temp_pos;
                break;
            }
            temp /= 10;
            temp_pos++;
        }
        if (foundPos == -1) {
            // Number not found
            System.out.println("No Magic is possible");
        } else {
            // Number found
            System.out.println("Special digit " + N + " is found in the " + foundPos + "th
position.");
            int sum = 0;
            temp = num;
            int i = 1;
            //loop for summation
            while (temp > 0) {
                int digit = temp % 10;
                if (foundPos % 2 == 0 && i % 2 == 0) {
                    sum += digit;
                } else if (foundPos % 2 != 0 && i % 2 != 0) {
                    sum -= digit;
                }
                temp /= 10;
                i++;
            }
            System.out.println("Magic number is : " + sum);
        }
    }
}
```

[SET-B]

SOLUTION

```
import java.util.Scanner;
public class Quiz2SetB {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Ancient Number: ");
        int num = sc.nextInt();
        System.out.print("Special digit: ");
        int N = sc.nextInt();

        int temp = num;
        int foundPos = -1;
        int temp_pos = 1;
        // Search the Number
        while (temp > 0) {
            int digit = temp % 10;
            if (digit == N) {
                foundPos = temp_pos;
                break;
            }
            temp /= 10;
            temp_pos++;
        }

        if (foundPos == -1) {
            // Number not found
            System.out.println("No Magic is possible");
        } else {
            // Number Found
            System.out.println("Special digit " + N + " is found in the " +
foundPos + "th position.");
            int product = 1;
            temp = num;
            int i = 1;
            //loop for multiplication
            while (temp > 0) {
                int digit = temp % 10;
                if (foundPos % 2 == 0 && i % 2 == 0) {
                    product *= digit;
                } else if (foundPos % 2 != 0 && i % 2 != 0) {
                    product *= digit;
                }
                temp /= 10;
                i++;
            }
            System.out.println("Magic number is : " + product);
        }
    }
}
```

[CSE110 Spring 2024 Lab Exam - 0X Tentative Solutions and Rubrics](#)

RUBRIC for Set A & B

SL	Points To Meet	Marks [10]
1.	Proper Syntax, import Scanner, Scanner Obj	1
2.	Take two user inputs & other variables	1.5
3.	Search the number and update position	2
4.	Number not found condition & print	1
5.	Loop → If number is found	1.5
6.	Proper conditions to check even/odd & update sum/product	3